Ritesh Somashekar

AI & ML Engineer

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PROFESSIONAL SUMMARY

- **AI & Machine Learning Engineer** with **5+ years** of experience building **LLM and RAG systems**, **NLP models**, and **enterprise data pipelines** across healthcare and public-safety domains. Published researcher (**ACM WSDM '26**, **IEEE PST 2026**) with hands-on expertise in **Llama 3, ChromaDB, Hugging Face, PyTorch, and Scikit-learn**, and in **hallucination detection, ontology-based validation, and LLM evaluation**.
- Strong background in **data engineering** with **Apache Airflow, Spark/PySpark, Snowflake, and Databricks**, and **cloud pipelines on AWS (S3, Lambda, Athena, Glue, SageMaker)** and **Azure**, with a focus on **HIPAA-compliant** healthcare data. Experienced in **MLOps** using **MLflow, Docker, Kubernetes, FastAPI,** and **CI/CD pipelines**.
- Demonstrated impact, including a **0.81 protocol F1 with a 3.0% hallucination rate** for RAG-generated scenarios, **99.9% pipeline reliability**, **74% higher data accuracy**, and **60% faster model training**. Adept at collaborating with researchers, public agencies, and business stakeholders to deliver **end-to-end AI solutions** from research to production.

TECHNICAL SKILLS

Programming & Scripting: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), SQL, R, Scala, Java, Bash

Generative AI & LLMs: Llama 3/3.1, Ollama, Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Vector Databases (ChromaDB, FAISS), Sentence Transformers, LoRA, Quantization-Aware Training (QAT)

LLM Evaluation & Safety: Hallucination Detection, Ontology-Based Validation, NLI-Based Validation, Semantic Similarity Scoring, Ablation Studies, Protocol F1, Confidence Scoring

Machine Learning & NLP: Supervised & Unsupervised Learning, Transformers, BERT, BioBERT, Hugging Face, XGBoost, LightGBM, Graph Analytics, Ensemble Models, Time Series Forecasting, Anomaly Detection, Feature Engineering, Model Explainability (SHAP)

MLOps & ML Infrastructure: MLflow, Docker, Kubernetes, Jenkins, CI/CD, FastAPI, Git, DVC, Model Deployment

Data Engineering & Big Data: Apache Airflow, Apache Spark, PySpark, Kafka, Databricks, Snowflake, Delta Lake, Hadoop, Hive, Sqoop, Informatica PowerCenter, Precisely Data360, SQL Server, MySQL, MongoDB

Cloud Platforms: AWS (S3, Lambda, Athena, Glue, SageMaker, SQS), Azure (Data Services, Stream Analytics)

Data Analysis & Visualization: Tableau, Power BI (DAX), Excel (VBA, Formulas), Statistical Analysis, Data Lineage

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA | May 2025 | GPA: 3.93/4.0

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | ***AI & ML Research Associate*** **Fairfax, VA | August 2024 - Present**

- Co-developed **SHIELD**, an online conversational AI training platform for 9-1-1 call-takers built with **Fairfax County DPSC**, using **LLM agents** to generate call scenarios from CAD data, deliver **real-time metacognitive nudges**, and produce post-call reflection analytics; published at **ACM WSDM '26**.
- **Led the data pipeline for both SHIELD and MATE-911** in **Python**, covering OCR parsing, abbreviation expansion, completeness checks, segmentation, standardization, and **ChromaDB** embedding of **160 gold-standard emergency scenarios** across accident, fire, medical, and alarm incidents.
- **Led the validation layer for both platforms**, building a **911 ontology validator** derived from **NENA** standards and DPSC procedures that combines rule-based and **LLM-based semantic checks** for unsafe content, protocol compliance, and **hallucination detection**, with PASS/REVIEW/FAIL confidence scoring and an **85% cosine-similarity** quality gate.
- Co-developed **MATE-911** (published at **IEEE PST 2026**), a multimodal **RAG** system using **Llama 3.1 8B (Ollama)**, **ChromaDB**, and **all-mpnet-base-v2** embeddings to generate protocol-grounded voice and text scenarios on commodity hardware, achieving **0.81 protocol F1**, a **3.0% hallucination rate**, a **97.6% pass rate**, and **sub-2-second latency**.
- Ran **ablation studies** showing RAG retrieval drives scenario quality (**F1 0.81 vs. 0.59** without it) and ontology validation acts as a safety guardrail, cutting hallucinations from **9.5% to 3.0%**.

- Built real-time **trace-log analytics** (event streams, Task Completion Rate, Average Field Latency, goal-alignment scoring) that feed an instructor **performance dashboard** for adaptive feedback.
- Applied **graph-based analytics** and **ensemble models** to large-scale disaster datasets on the **ORC Hopper HPC cluster** to forecast community coping trends with an **RMSE of 0.89**, visualizing findings in **R** and **Tableau**.
- Implemented **model explainability** using **SHAP** for **intent classification** and **behavioral anomaly detection** models, surfacing the features driving each prediction to guide threshold tuning and model governance; the **LightGBM** anomaly detector with threshold-based alerting reduced false positives by **25%**.
- Fine-tuned **Llama 3** and transformer models using **LoRA/PEFT** and **Quantization-Aware Training (QAT)**, reducing model size by **40%** while keeping accuracy within **3%** of full-precision benchmarks.
- Established **MLflow** experiment tracking and model registry for LLM and RAG experiments, logging parameters, prompts, and evaluation metrics to make every run reproducible and comparable across model versions.
- Took models from **proof of concept to production** by serving them through **FastAPI** and **Docker** with **CI/CD**, reducing release cycles from **2 weeks to 5 days** and enabling **A/B testing** and pre/post-deployment evaluation of model versions.

Carelon Global Solutions (Elevance Health) | ***Machine Learning Engineer*** **Bengaluru, India | May 2022 - August 2023**

- Designed and managed **20+ enterprise ETL pipelines**, boosting data throughput by **70%** and ensuring high availability for downstream AI/BI applications.
- Automated workflows with **Apache Airflow, Python,** and **AWS (S3, Lambda, Athena)**, achieving **99.9% pipeline reliability** with end-to-end lineage tracking.
- Built a daily ingestion workflow from **NPPES** repositories, improving data accuracy by **74%** and cutting processing overhead by **60%**.
- Engineered ingestion pipelines for **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating model training by **60%** and reducing preprocessing time by **70%**.
- Built NLP models using **BERT** and **BioBERT** for clinical entity extraction from unstructured medical text, achieving **80% accuracy** and reducing processing time by **55%**.
- Built scalable **PySpark** ETL on **AWS (Glue, SageMaker)** with **Delta Lake**, improving data reliability by **30%** under **HIPAA**, and improved **Scikit-learn/XGBoost** models, increasing accuracy by **8%** and cutting training time by **15%**.
- Containerized ML inference services with **Docker, Kubernetes,** and **Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- Delivered self-service BI pipelines integrating **Tableau, Power BI,** and SQL-based models, reducing reporting latency by **40%** and enabling anomaly detection for healthcare analytics.

Carelon Global Solutions (Elevance Health) | ***Data Engineer*** **Bengaluru, India | October 2020 - April 2022**

- Partnered with stakeholders on the **Seven Plus Locations** project to align ETL pipelines with business KPIs, improving claims data accuracy and driving cost savings.
- Redesigned **claims-balancing ETL workflows**, improving provider insights and reducing operational overhead by **60%**.
- Automated **medical invoice processing** pipelines, cutting manual errors by **70%** and improving the reliability of financial reporting.
- Migrated large-scale processing workloads from **Hive** to **Apache Spark/PySpark**, and moved **Sqoop/Hadoop** ingestion to **Azure Data Services**, improving performance, scalability, and pipeline efficiency.
- Developed **RESTful microservices** with **OAuth 2.0** and API versioning, and designed **MongoDB** data models with **Python/Java** validation scripts (UDFs) to improve data quality and consistency.

PUBLICATIONS

- R. S. Nayaka, **R. Somashekar**, K. B. Laskey, L. Wells II, and H. Purohit. "SHIELD: A Framework for Online Conversational AI System for Self-Regulated Learning at Emergency Communications Centers." In *Proceedings of the 19th ACM International Conference on Web Search and Data Mining (WSDM '26)*, Boise, ID, 2026. doi:10.1145/3773966.3779411
- R. S. Nayaka, **R. Somashekar**, and H. Purohit. "MATE-911: Multimodal AI-powered Training Environment for Next-Generation 911 Dispatcher Preparedness." In *IEEE PST Conference*, 2026.